UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 002 · Solutions of homogeneous DE: complementary functions · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Unit 2 deck 002, slide 7 (rendered .pptx)
T002 00:09:10	Unit 2 AV-summary transcript 002 (“Rules for finding the Complementary Function, Problems”), segment start
CW U2 P8	Unit 2 class-work problem 8
ESA Jan-24 s4 2c	Past paper: UE23MA141A Jan 2024 set 4, question 2.c

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2. 1 · The auxiliary equation and the complementary function
3. 2 · Second-order equations: three cases
4. 3 · Equations of order n: combining the cases
5. Part B · Symbol glossary

A Formula sheet

1 The auxiliary equation and the complementary function

E1.1 · Trial solution and its derivatives ESA ★★

$$y = e^{mx}, \qquad y' = m\,e^{mx}, \qquad y'' = m^2 e^{mx}$$

GIVES	Turns $f(D)y = 0$ into $f(m)e^{mx} = 0$.
APPLIES	Any linear DE with constant coefficients.
NOTE	$D^k e^{mx} = m^k e^{mx}$, so $f(D)e^{mx} = f(m)e^{mx}$.

S002·4 · T002 00:02:04

E1.2 · Auxiliary equation ESA ★★★

$$f(D)y = 0 \implies f(m) = m^n + k_1 m^{n-1} + \cdots + k_n = 0$$

GIVES The n roots $m_1, \ldots, m_n$ that fix the CF.

APPLIES Homogeneous, constant coefficients.

NOTE Replace D by m (equivalently $y \to 1$, $y' \to m$, $y'' \to m^2$, $y''' \to m^3$).

S002·5 · S002·10 · T002 00:03:18 · T002 00:07:40

2 Second-order equations: three cases

E2.1 · Case I: real and distinct roots ESA ★★★

$$m_1 \neq m_2 \text{ real}: \quad y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

GIVES The CF.

APPLIES Discriminant $k_1^2 - 4k_2 > 0$.

NOTE Slide 002·7: $m = -3, 1$ gives $y = c_1 e^{-3x} + c_2 e^{x}$.

S002·7 · T002 00:05:40

E2.2 · Case II: real and equal roots ESA ★★★

$$m_1 = m_2 = m: \quad y = (c_1 + c_2 x) e^{mx}$$

GIVES The CF.

APPLIES $k_1^2 - 4k_2 = 0$, $m = -\frac{k_1}{2}$.

NOTE "This is the formula you should remember" (T002 00:09:10).

S002·8 · T002 00:08:57 · T002 00:09:10

E2.3 · Case III: complex roots ESA ★★★

$$m = \alpha \pm i\beta: \quad y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x)$$

GIVES The CF, in real form.

APPLIES $k_1^2 - 4k_2 < 0$; α = real part, β = imaginary part ($\beta > 0$).

NOTE $\alpha = 0$ (roots $\pm i\beta$) gives $c_1 \cos \beta x + c_2 \sin \beta x$.

S002·9 · T002 00:10:56

E2.4 · Roots of the quadratic AE

ESA ★★

$$m = \frac{-k_1 \pm \sqrt{k_1^2 - 4k_2}}{2}$$

GIVES Which case applies, from the sign of the discriminant.

APPLIES AE $m^2 + k_1 m + k_2 = 0$.

NOTE Lecture: "$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$" with $a = 1, b = -4, c = 13$ for slide 002·9.

ADDED FOR CLARITY T002 00:11:56

3 Equations of order n: combining the cases

E3.1 · Case (i): n real distinct roots

ESA ★★★

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}$$

GIVES The CF of an n-th order equation.

APPLIES All roots real and different.

NOTE n constants.

S002·11 · T002 00:14:12

E3.2 · Case (ii): a root repeated k times

ESA ★★★

$$y = (c_1 + c_2 x + c_3 x^2 + \cdots + c_k x^{k-1}) e^{mx}$$

GIVES The block of the CF for a k-fold real root m.

APPLIES Slide 002·12 has $k = n$ (all roots equal) and $k = 4$.

NOTE Stop at x^{k-1}: a double root gives $c_1 + c_2 x$, a triple root $c_1 + c_2 x + c_3 x^2$.

S002·12 · T002 00:14:58 · T002 00:15:15

E3.3 · Case (iii): equal roots with distinct ones

ESA ★★

$$y = (c_1 + c_2 x + c_3 x^2 + c_4 x^3) e^{mx} + c_5 e^{m_5 x} + \cdots + c_n e^{m_n x}$$

GIVES The CF when four roots are equal and the rest distinct.

APPLIES $m_1 = m_2 = m_3 = m_4 = m$.

NOTE Add the blocks of each root.

S002·12 · T002 00:16:12 · T002 00:16:37

E3.4 · Case (iv): two pairs of complex roots ESA ★★

$$y = e^{\alpha x}(c_1 \cos \beta x + c_2 \sin \beta x) + e^{\gamma x}(c_3 \cos \delta x + c_4 \sin \delta x) + c_5 e^{m_5 x} + \dots$$

GIVES The CF.

APPLIES Roots $\alpha \pm i\beta$, $\gamma \pm i\delta$, rest real and distinct.

NOTE Each pair has its own two constants (CW P5).

S002·13 · T002 00:17:08 · T002 00:17:36

E3.5 · Case (v): a complex pair occurring twice ESA ★★

$$y = e^{\alpha x}\left[(c_1 + c_2 x)\cos \beta x + (c_3 + c_4 x)\sin \beta x\right] + c_5 e^{m_5 x} + \dots$$

GIVES The CF.

APPLIES $\alpha \pm i\beta$ repeated twice.

NOTE Both the cosine and the sine get a polynomial of degree 1 (CW P7).

S002·13 · T002 00:18:21 · T002 00:18:51

E3.6 · A complex pair occurring three times ESA ★

$$y = e^{\alpha x}\left[(c_1 + c_2 x + c_3 x^2)\cos \beta x + (c_4 + c_5 x + c_6 x^2)\sin \beta x\right] + \dots$$

GIVES The CF.

APPLIES $\alpha \pm i\beta$ repeated three times.

NOTE Said in the lecture; not on the slides.

TRANSCRIPT ONLY T002 00:19:14 · T002 00:19:32 · T002 00:19:45

E3.7 · Zero root ESA ★★

$$m = 0: \quad c\,e^{0x} = c$$

GIVES A constant term in the CF.

APPLIES $f(m)$ has a factor m (no y term in the DE).

NOTE Slide 002·14: $m = 0, \pm 2i$ gives $c_1 + c_2 \cos 2x + c_3 \sin 2x$. A double zero root gives $c_1 + c_2 x$.

S002·14 · T002 00:21:20

B Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$; $D^2 = \frac{d^2}{dx^2}$	E1.2
$f(D)$	Linear differential operator	$D^n + k_1 D^{n-1} + \cdots + k_n$	E1.2
$f(m)$	Auxiliary polynomial	$f(D)$ with D replaced by m	E1.2
$m,\ m_1, \ldots, m_n$	Roots of the AE	Exponents of the trial solution e^{mx}	E2.1–E3.5
$k_1, \ldots, k_n$	Constant coefficients	Given numbers in the DE	E1.2, E2.4
$c_1, \ldots, c_n$	Arbitrary constants	n of them for order n; fixed by initial conditions	all
$\alpha,\ \beta$	Real and imaginary parts	Of a complex root $\alpha \pm i\beta$ ($\beta > 0$)	E2.3, E3.4, E3.5
$\gamma,\ \delta$	Second complex pair	$\gamma \pm i\delta$ in case (iv)	E3.4
Δ	Discriminant	$k_1^2 - 4k_2$ (added for clarity)	F2.1
k	Multiplicity	Times a root is repeated. Not the coefficients k_1, k_2	E3.2
$\theta,\ z$	Legendre/Cauchy substitution	$z = \log(ax + b)$, $\theta = \frac{d}{dz}$ (classes 011; table C4.2 only)	C4.2

Remark. The slides use $k_1, k_2, \ldots$ for the coefficients; this pack also uses k alone for the multiplicity of a root (how often it repeats). The two are different. The discriminant Δ is added for clarity. θ and z appear only in table C4.2, for the Legendre and Cauchy questions of class 011.